Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

V SEMESTER B. TECH (MECHANICAL/IP ENGG.) END SEMESTER EXAMINATIONS, DECEMBER 2019

SUBJECT: METROLOGY AND MEASUREMENTS [MME 3104]

REVISED CREDIT SYSTEM

Time: 3 Hours MAX. MARKS: 50

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Draw neat sketch of Bourdon Tube Pressure Gauge with electrical read-out and show its generalized measurement system block diagram with various functional elements. **3**

1B. Define and derive an expression for gauge factor. **3**

1C. Explain with neat sketch the calibration of strain gauge using shunt resistance and derive the expression for strain. Also calculate the strain undergone by a strain gauge having gauge factor 2 and initial resistance of 200Ω. The change in resistance is 0.016Ω. **4**

2A. With the help of a neat sketch explain the working of a Hydraulic Load Cell. **3**

2B. With the help of a neat sketch explain the working of a Rope brake drum dynamometer and derive the expression for Brake Power related to it. **3**

2C. With the help of neat sketch explain the working of Bridgeman gauge and derive the expression for unknown pressure. **4**

3A. Give details of M112 slip gauge set. Wring a dimension of 64.3485 using this set. Also show the setup schematically. **3**

3B. Draw neat sketch of taper plug gauge and explain the procedure of checking a taper hole with it. **3**

3C. Design a plug and ring gauge for the fit $\phi 32E_8d_5$. Refer table 1 for data. **4**

4A. Explain Periodic and Progressive pitch errors usually found in screw threads in detail. **3**

4B. With the help of neat sketch derive the expression for effective diameter using 2-wire method. **3**

4C. Explain with neat sketch the procedure for flatness measurement of a surface plate. **4**

5A. With the help of neat sketch and schematic layout, explain the working of Taylor Hobson Talysurf. **5**

5B. A M 20 x 2.5 plug screw gauge is checked for effective diameter by a floating carriage micrometer with best size wire and the following readings were noted: **3**
Diameter of standard cylinder = 18.001mm.
Micrometer reading over standard cylinder with two wires of same diameter = 14.6420mm.
Micrometer reading over the plug screw gauge with two wires of same diameter = 14.2616mm.
Calculate the effective diameter of the gauge

5C. Calculate the CLA (Ra) value of a surface for which the sampling length was 0.6 mm. The graph was drawn to a vertical magnification of 5000 and a horizontal magnification of 200 and the areas above and below the datum line were **2**

Above :	200	120	160	130	mm^2
Below :	30	40	80	60	mm^2

Table1 for Question No. 3C

Basic Size (mm)	
Above	**Up to and including**
6	10
10	18
18	30
30	50
50	80
80	120

Tolerance Grades	**IT5**	**IT6**	**IT7**	**IT8**	**IOT9**	**IT10**
Xi (µm)	**7i**	**10i**	**16i**	**25i**	**40i**	**64i**

Type	**Fundamental Deviation (µm)**
D	$\mathbf{16D^{0.44}}$
E	$\mathbf{11D^{0.41}}$
F	$\mathbf{5.5D^{0.41}}$
G	$\mathbf{2.5D^{0.34}}$